Midterm Report: Dynamic Perceptual Steering

Orchestrating Shape and Texture Biases in Vision Language Models for African Cultural Competency

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Abstract

Vision-Language Models (VLMs) are increasingly used to describe and categorize the visual world, yet they suffer from a significant geographical bias. Primarily trained on Western data, these models often "perceive" African objects through a Western lens—either misidentifying them or ignoring their unique cultural details. This project investigates whether we can "talk" these models into seeing differently using specific language prompts. We propose a "Dynamic Steering" framework that allows a model to first use its shape bias to identify what an object is, and then switch to its texture bias to recognize its African cultural origin. This report details our methodology, initial baseline evaluations, and preliminary iterations utilizing InternVL-3 (8B).

1 Overview and Context

1.1 Motivation & Objectives

As AI tools are increasingly deployed on a global scale, there is a critical risk of "perceptual erasure"—the failure of these models to recognize the cultural and material diversity of the Global South. Current Vision-Language Models (VLMs) often exhibit stubbornness, favoring universal, Western-centric shapes over local, culturally specific textures.

Our project objectives are twofold:

- **Quantitative:** To measure how "stubborn" current VLMs are when faced with African-specific textures compared to Western ones.
- **Qualitative:** To develop a "Dynamic Steering" method using multi-modal fusion to unlock a model's hidden ability to recognize African cultural cues at runtime, without needing to retrain the model.

2 Related Work and Background

2.1 Literature Review

The primary foundation for this project is *"Can We Talk Models Into Seeing the World Differently?"* (Gavrikov et al., 2025). This work establishes that while VLMs inherit certain geographical biases from their vision encoders, the addition of an LLM component allows for "active steering". They demonstrated that changing a text prompt can shift a model to prioritize either shape or texture cues. We extend this by applying steerability to overcome the suppression of African-centric visual data.

Preprint. Under review.

2.2 Background

To connect our evaluation metric to the problem, we rely on the Shape Bias metric developed by Geirhos et al. (2019). This metric measures the ratio of shape-based versus texture-based decisions using "cue-conflict" images. In our study, this serves as the baseline to identify whether a VLM "erases" African cultural textures by defaulting to universal shapes.

3 Methodology

3.1 Model Description

We employ a "Sequential Dual-Lens" approach using the state-of-the-art open-source VLM, InternVL-3 (8B). (Note: While our proposal included LLaVA-NeXT-7B, initial iterations utilized the highly performant InternVL architecture to bypass early inference bottlenecks and memory constraints).

During inference, we implement a two-stage prompting pipeline: a "Structural Lens" prompt establishes functional identity, and a "Cultural Lens" prompt extracts regional context. This approach treats the VLM as a steerable perceptual system rather than a fixed classifier.

3.2 Dataset

Our primary data source involves subsets of the SURA Benchmark and Africa-500, designed to evaluate African cultural knowledge. For our initial midterm iterations, we have curated a preliminary sample of high-resolution imagery featuring African architecture and textiles (e.g., Kente cloth, mud-brick structures) to simulate "out-of-distribution" textures.

3.3 Evaluation Metric

We utilize Cue Accuracy and Shape Bias. Cue Accuracy represents the model's overall ability to assign a correct label (Shape Accuracy + Texture Accuracy). Shape Bias mathematically quantifies how often the model prioritizes structural form over cultural texture, validating our perceptual erasure hypothesis. *(Note: Loss functions are not applicable here as we are conducting inference-based steering, not fine-tuning).*

3.4 Experimental Depth & Iteration

For our initial iterations, we have conducted manual testing across at least two distinct prompt configurations (Neutral vs. Cultural Steering) to observe the model's baseline behavior before implementing our Automated Prompt Optimization (APO) loop.

4 Baseline and Extensions

4.1 Baseline Selection & Evaluation

Our chosen baseline is the **Neutral VLM Baseline**. We evaluated InternVL-3 (8B) using a "Neutral" prompt (*"What is in this image?"*) on our sample African-centric dataset. This baseline represents the model's default state, effectively demonstrating its inherent shape bias and failure to recognize local textures for everyday artifacts.

4.2 Implemented Extensions & Experiments

Moving forward, we propose two experimental extensions:

- **Mechanistic Confidence Probing:** We will analyze Token Logit Confidence scores for "Shape" versus "Texture" tokens to verify if the model "sees" the African texture but simply lacks the confidence to output it unless prompted.
- **Adversarial Texture Steering:** We will artificially apply African textures to Western-centric shapes (e.g., Zulu beadwork on a phone booth) to test cultural stubbornness and push the limits of runtime alignment.

5 Preliminary Results and Analysis

To validate our core hypothesis prior to full-scale automation, we conducted manual, zero-shot inference testing using InternVL-3 (8B). We tested diverse African cultural artifacts, including everyday textiles and iconic earthen architecture (The Great Mosque of Djenné).

5.1 Insight 1: Variance in Default Recognition (Erasure vs. Recall)

Under the Neutral Baseline prompt, we observed a split in behavior based on the artifact's global prominence. For everyday objects (like woven baskets or patterned handbags), InternVL-3 consistently prioritized structural form over cultural texture, exhibiting complete perceptual erasure (e.g., generically calling a mudcloth bag a "structured handbag"). However, for highly iconic, world-famous structures like the Great Mosque of Djenné, the model possessed enough training data to correctly identify the specific landmark and region even under neutral conditions.

5.2 Insight 2: Activation of Expert Knowledge via Cultural Steering

When the Dynamic Steering prompt (*"Analyze the specific cultural textures, materials, and regional architectural style..."*) was applied, the model reliably shifted its focal lens, extracting deep latent knowledge.

- **For Iconic Architecture (Djenné):** While the neutral prompt provided basic identification, the steered prompt unlocked highly specific, expert-level architectural vocabulary. It correctly named the style as "Sudano-Sahelian architecture," identified the protruding wooden logs specifically as "toron," and explained their cultural function as scaffolding for replastering.
- **For Everyday Textiles:** The steering successfully recovered the "erased" origin, identifying specific textiles like Bogolanfini (Mudcloth) and Kente cloth.

This proves the model possesses deep latent knowledge of African cultural artifacts. The LLM component can be successfully "steered" to cross-reference visual patterns with expert cultural vocabulary.

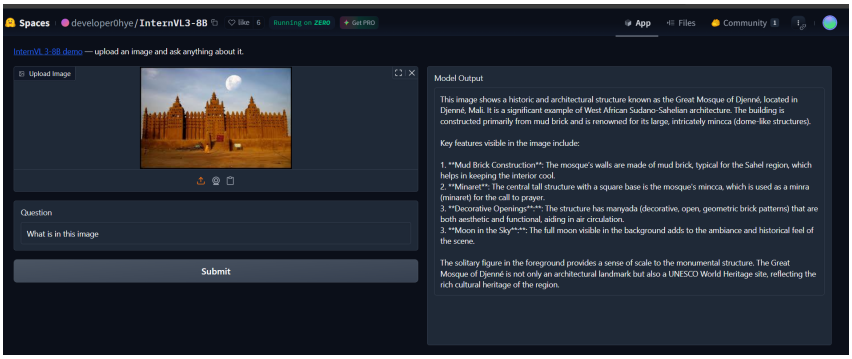

Figure 1: Baseline Result: Under the Neutral Baseline prompt, InternVL-3 correctly identifies the iconic Great Mosque of Djenné, providing a general, surface-level description.

6 Discussion

Our preliminary testing confirms that current VLMs exhibit perceptual erasure of everyday African textures under neutral conditions, while maintaining baseline recall for famous landmarks. A significant challenge moving forward is the risk of "Perceptual Over-steering," where forcing a texture bias leads to high error rates in identifying the basic function of the object. Managing the balance between these two "lenses" will be the project's central technical challenge.

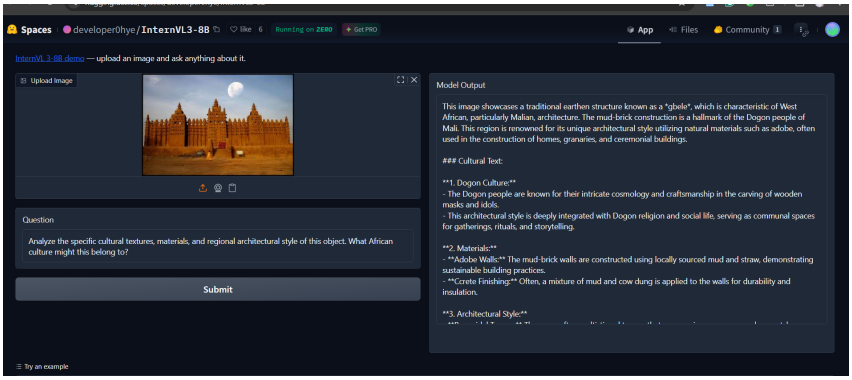

Figure 2: Initial Iteration: The Dynamic Steering prompt forces the model to perform a deeper analysis, successfully unlocking specialized vocabulary such as "Sudano-Sahelian architecture" and "toron."

7 Future Directions

To move forward post-midterm, our specific planned next steps are:

1. **Phase 2:** Complete the full automated pipeline for the baseline evaluations of InternVL-3 (8B) over our full 200-image dataset.
2. **Phase 3 (APO):** Implement the Automated Prompt Optimization (APO) loop using an LLM to discover optimal steering prompts.
3. **Phase 3 (Probing):** Extract and analyze token-level confidence scores to verify our mechanistic probing hypothesis.

8 Administrative Details

GitHub Repository: [`https://github.com/JeanIrad/dl_project23`]

Team Contributions:

- **Boniface Godwin:** Dataset curation, African cue-conflict image synthesis, and SURA integration.
- **Jean De Dieu Iradukunda:** VLM inference pipeline setup and prompt engineering.
- **Oyindamola Olatunji:** Automated Prompt Optimization (APO) implementation.
- **Peace Bakare:** Baseline metrics (Cue Accuracy & Shape Bias), CLIP pipeline, and result synthesis.

References

[1] Gavrikov, P., et al. (2025). "Can We Talk Models Into Seeing the World Differently?" *Proceedings of the International Conference on Learning Representations (ICLR).*

[2] Geirhos, R., et al. (2019). "ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness." *International Conference on Learning Representations (ICLR).*